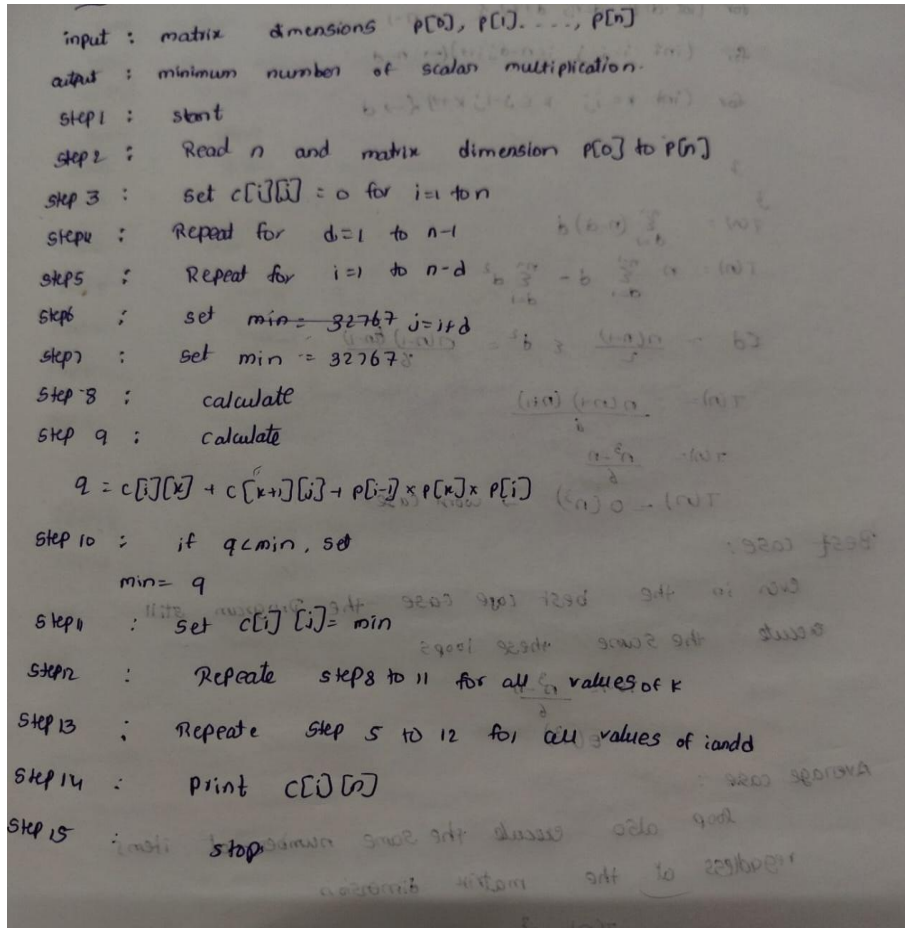


Practical- 6

Aim: To Implementation of chain matrix multiplication using dynamic programming.

chain matrix multiplication

Algorithm:



input : matrix dimensions $p[0], p[1], \dots, p[n]$
 output : minimum number of scalar multiplication.
 step 1 : start
 step 2 : Read n and matrix dimension $p[0]$ to $p[n]$
 step 3 : set $c[i][i] = 0$ for $i = 1$ to n
 step 4 : Repeat for $d = 1$ to $n-1$
 step 5 : Repeat for $i = 1$ to $n-d$
 step 6 : set $min = 32767$ $j = i+d$
 step 7 : set $min = 32767$
 step 8 : calculate
 step 9 : calculate

$$q = c[i][k] + c[k+1][j] + p[i] \times p[k] \times p[j]$$

 step 10 : if $q < min$, set
 $min = q$
 step 11 : set $c[i][j] = min$
 step 12 : Repeat steps 8 to 11 for all values of k
 step 13 : Repeat step 5 to 12 for all values of i and d
 step 14 : print $c[i][n]$
 step 15 : stop

Program:

```
#include <iostream>
```

```
using namespace std;
```

```
class MatrixChain
```

```
{
```

```
private:
```

```
int p[20];
```

```
int c[20][20];
```

```
int n;
```

```
public:
```

```
void getInput()
```

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```
{
    cout << "Enter number of matrices: ";
    cin >> n;
    cout << "Enter dimensions: ";
    for (int i = 0; i <= n; i++)
    {
        cin >> p[i];
    }
}

void matrixChain()
{
    for (int i = 1; i <= n; i++)
    {
        c[i][i] = 0;
    }
    for (int d = 1; d < n - 1 + 1; d++)
    {
        for (int i = 1; i <= n - d; i++)
        {
            int j = i + d;
            int min = 32767;

            for (int k = i; k <= j - 1; k++)
            {
                int q = c[i][k] + c[k + 1][j]
                    + p[i - 1] * p[k] * p[j];

                if (q < min)
                {
                    min = q;
                }
            }
        }
    }
}
```



```
        c[i][j] = min;
    }
}

cout << "\nMinimum number of scalar multiplications = "
    << c[1][n] << endl;
}

void display()
{
    cout << "\nDP Table:\n";

    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            if (j < i)
                cout << "0 ";
            else
                cout << c[i][j] << " ";
        }
        cout << endl;
    }
}

};

int main()
{
    MatrixChain obj;
    obj.getInput();
    obj.matrixChain();
    obj.display();
    return 0;
}
```

Output:

```
PS C:\MinGW\bin> & "C:\Users\chinn\Downloads\dsa in c++\practical-6\chain matrix.exe"
Enter number of matrices: 4
Enter dimensions: 3
2
4
2
5

Minimum number of scalar multiplications = 58

DP Table:
0 24 28 58
0 0 16 36
0 0 0 40
0 0 0 0
```

Time complexity:

for (int d = 0; d < n; d++) {
 for (int i = 0; i < n - d; i++) {
 for (int j = i; j < n - d; j++) {
 // ...
 }
 }
}

$$T(n) = \sum_{d=0}^{n-1} (n-d)d$$
$$T(n) = n \sum_{d=1}^{n-1} d - \sum_{d=1}^{n-1} d^2$$
$$= \frac{n(n-1)}{2} - \frac{(n-1)n(2n-1)}{6}$$
$$T(n) = \frac{n(n-1)(n+1)}{6}$$
$$T(n) = \frac{n^3 - n}{6}$$
$$T(n) = O(n^3) \rightarrow \text{worst case.}$$

Best case:

even in the best case the program still execute the same these loops

$$T(n) = \frac{n^3 - n}{6}$$

Time complexity: $O(n^3)$

Average case:

loop also execute the same number of items regardless of the matrix dimension

$$T(n) = \frac{n^3 - n}{6}$$
$$T(n) = O(n^3)$$



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Conclusion: The Chain Matrix Multiplication problem was successfully implemented using Dynamic Programming. The algorithm finds the optimal order of matrix multiplication and minimizes the number of scalar multiplications. It stores the solutions of smaller subproblems in a DP table and uses them to solve the complete problem efficiently.